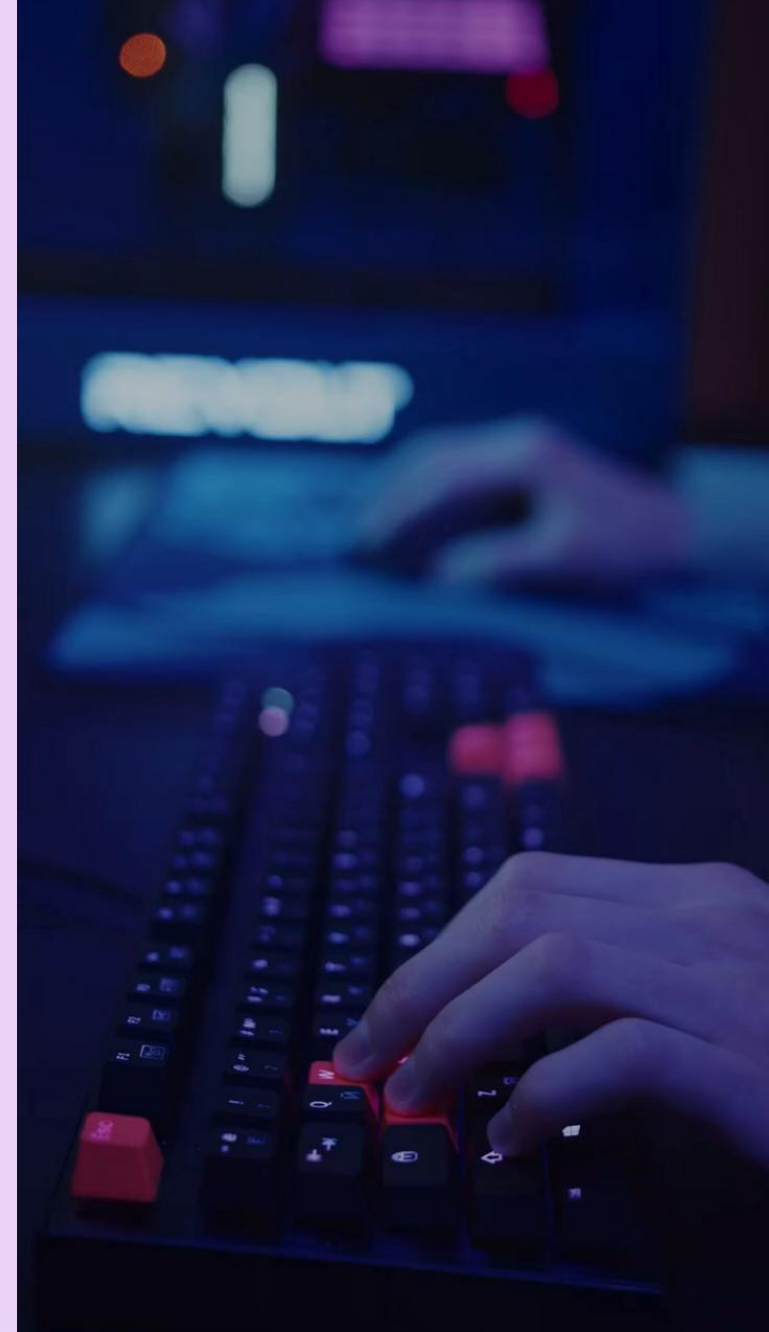


COMPUTER DATA REPRESENTATION

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Computer Data Representation

Computers do not understand letters, words, or symbols the way humans do. Instead, they operate using **binary (0s and 1s)**. Data representation refers to the way different types of data — such as numbers, text, images, audio, and video — are **encoded and stored** inside a computer's memory.

What Computer Data Representation?

Data Representation is the method of storing, processing, and transmitting information in a form that computers can understand.

Since computers use the **binary number system (base-2)**, all types of data are ultimately represented as combinations of:

0 → OFF (no electrical signal)

1 → ON (electrical signal present)

Number Representation

Binary Number System (Base-2)

- Uses digits 0 and 1
- Each digit is called a **bit** (binary digit)
- Group of 8 bits = **1 byte**

Example:

Decimal 13 → Binary 1101

Other Number System

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Other Number System

Number System	Base	Digits Used	Example
Binary	2	0, 1	1010
Octal	8	0–7	12 (decimal 10)
Decimal	10	0–9	45
Hexadecimal	16	0–9, A–F	A3 (decimal 163)

Decimal Number System

Decimal number system is a base 10 number system having 10 digits from 0 to 9. This means that any numerical quantity can be represented using these 10 digits. Decimal number system is also a positional value system.

Decimal Number System

This means that the value of digits will depend on its position. Let us take an example to understand this. Say we have three numbers – 734, 971 and 207. The value of 7 in all three numbers is different–

- In 734, value of 7 is 7 hundreds or 700 or 7×100 or 7×10^2
- In 971, value of 7 is 7 tens or 70 or 7×10 or 7×10^1
- In 207, value of 7 is 7 units or 7 or 7×1 or 7×10^0

Binary Number System

The easiest way to vary instructions through electric signals is two-state system – on and off. On is represented as 1 and off as 0, though 0 is not actually no signal but signal at a lower voltage. The number system having just these two digits – 0 and 1 – is called binary number system.

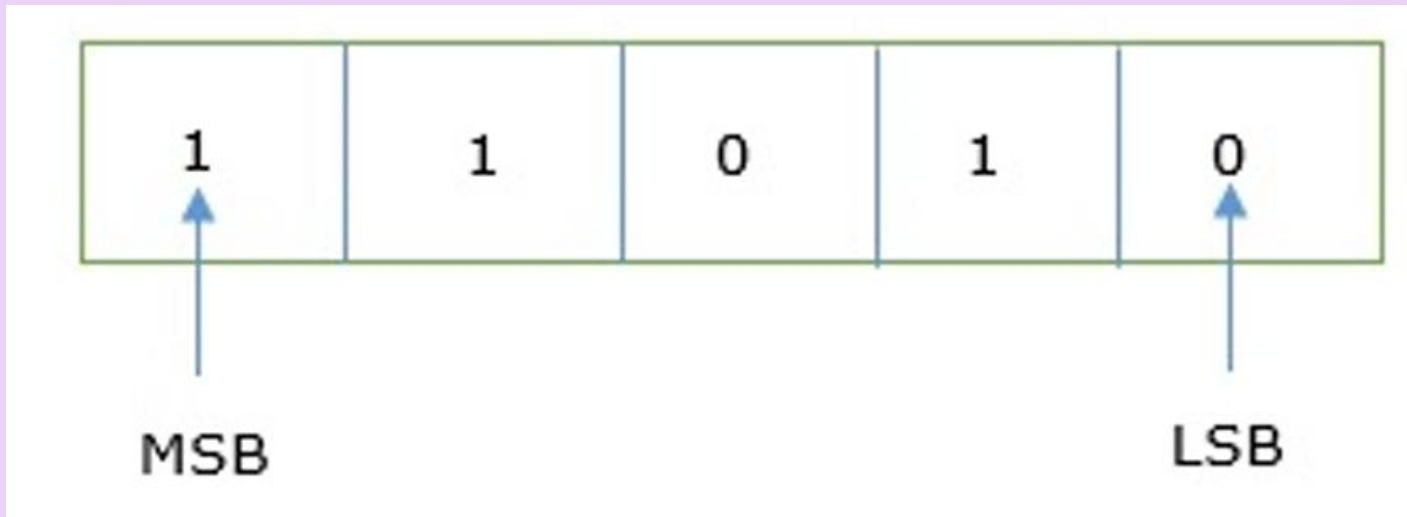
Binary Number System

Each binary digit is also called a bit. Binary number system is also positional value system, where each digit has a value expressed in powers of 2, as displayed here.

2^5	2^4	2^3	2^2	2^1	2^0
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Binary Number System

In any binary number, the rightmost digit is called least significant bit (LSB) and leftmost digit is called most significant bit (MSB).



Binary Number System

And decimal equivalent of this number is sum of product of each digit with its positional value.

$$\begin{aligned}11010_2 &= 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 0 \times 2^0 \\&= 16 + 8 + 0 + 2 + 0 \\&= 26_{10}\end{aligned}$$

Binary Number System

Computer memory is measured in terms of how many bits it can store. Here is a chart for memory capacity conversion.

- 1 byte (B) = 8 bits
- 1 Kilobytes (KB) = 1024 bytes
- 1 Megabyte (MB) = 1024 KB
- 1 Gigabyte (GB) = 1024 MB
- 1 Terabyte (TB) = 1024 GB
- 1 Exabyte (EB) = 1024 PB
- 1 Zettabyte = 1024 EB
- 1 Yottabyte (YB) = 1024 ZB

Octal Number System

Octal number system has eight digits – 0, 1, 2, 3, 4, 5, 6 and 7. Octal number system is also a positional value system with where each digit has its value expressed in powers of 8, as shown here –

8^5	8^4	8^3	8^2	8^1	8^0
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Octal Number System

Decimal equivalent of any octal number is sum of product of each digit with its positional value.

$$\begin{aligned} 726_8 &= 7 \times 8^2 + 2 \times 8^1 + 6 \times 8^0 \\ &= 448 + 16 + 6 \\ &= 470_{10} \end{aligned}$$

Hexadecimal Number System

Hexadecimal number system has 16 symbols – 0 to 9 and A to F where A is equal to 10, B is equal to 11 and so on till F. Hexadecimal number system is also a positional value system with where each digit has its value expressed in powers of 16, as shown here –

16^5	16^4	16^3	16^2	16^1	16^0
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Hexadecimal Number System

Decimal equivalent of any hexadecimal number is sum of product of each digit with its positional value.

$$\begin{aligned} 27FB_{16} &= 2 \times 16^3 + 7 \times 16^2 + 15 \times 16^1 + 10 \times 16^0 \\ &= 8192 + 1792 + 240 + 10 \\ &= 10234_{10} \end{aligned}$$

Number System Relationship

The following table depicts the relationship between decimal, binary, octal and hexadecimal number systems.

HEXADECIMAL	DECIMAL	OCTAL	BINARY
0	0	0	0000
1	1	1	0001
2	2	2	0010
3	3	3	0011
4	4	4	0100
5	5	5	0101
6	6	6	0110

Number System Relationship

The following table depicts the relationship between decimal, binary, octal and hexadecimal number systems.

7	7	7	0111
8	8	10	1000
9	9	11	1001
A	10	12	1010
B	11	13	1011
C	12	14	1100
D	13	15	1101
E	14	16	1110
F	15	17	1111

ASCII (American Standard Code for Information Interchange)

Besides numerical data, computer must be able to handle alphabets, punctuation marks, mathematical operators, special symbols, etc. that form the complete character set of English language. The complete set of characters or symbols are called alphanumeric codes. The complete alphanumeric code typically includes –

- 26 upper case letters
- 26 lower case letters
- 10 digits
- 7 punctuation marks
- 20 to 40 special characters

ASCII Code - Character to Binary

0	0011 0000	I	0100 1001	b	0110 0010	v	0111 0110
1	0011 0001	J	0100 1010	c	0110 0011	w	0111 0111
2	0011 0010	K	0100 1011	d	0110 0100	x	0111 1000
3	0011 0011	L	0100 1100	e	0110 0101	y	0111 1001
4	0011 0100	M	0100 1101	f	0110 0110	z	0111 1010
5	0011 0101	N	0100 1110	g	0110 0110		
6	0011 0110	O	0100 1111	h	0110 1000	:	0011 1010
7	0011 0110	P	0101 0000	i	0110 1001	;	0011 1011
8	0011 1000	Q	0101 0001	j	0110 1010	?	0011 1111
9	0011 1001	R	0101 0010	k	0110 1011	.	0010 1110
		S	0101 0011	l	0110 1100	,	0010 1111
		T	0101 0100	m	0110 1101	!	0010 0001

A	0100 0001	T	0101 0100	m	0110 1101	!	0010 0001
B	0100 0010	U	0101 0101	n	0110 1110	,	0010 1100
C	0100 0011	V	0101 0110	o	0110 1111	"	0010 0010
D	0100 0100	W	0101 0111	p	0111 0000	(	0010 1000
E	0100 0101	X	0101 1000	q	0111 0001	)	0010 1001
F	0100 0110	Y	0101 1001	r	0111 0010	space	0010 0000
G	0100 0111	Z	0101 1010	s	0111 0011		
H	0100 1000			t	0111 0100		
		a	0110 0001	u	0111 0101		



THANK YOU!

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